

Valuation fields

1 Absolute values and completion

Definition 1. Let $\mathbf{k}$ be a field. An *absolute value* on $\mathbf{k}$ is a function $|\cdot| : \mathbf{k} \rightarrow \mathbb{R}_{\geq 0}$ satisfying the following properties for all $x, y \in \mathbf{k}$:

- (a) $|x| = 0$ if and only if $x = 0$;
- (b) $|xy| = |x| \cdot |y|$;
- (c) $|x + y| \leq |x| + |y|$.

A field $\mathbf{k}$ equipped with an absolute value $|\cdot|$ is called a *valuation field*.

Remark 2. Let $\mathbf{k}$ be a field. Recall that a (additive) valuation on $\mathbf{k}$ is a function $v : \mathbf{k}^\times \rightarrow \mathbb{R}$ such that

- $\forall x, y \in \mathbf{k}^\times, v(xy) = v(x) + v(y)$;
- $\forall x, y \in \mathbf{k}^\times, v(x + y) \geq \min\{v(x), v(y)\}$.

We can extend v to the whole field $\mathbf{k}$ by defining $v(0) = +\infty$. Fix a real number $\varepsilon \in (0, 1)$. Then v induces an absolute value $|\cdot|_v : \mathbf{k} \rightarrow \mathbb{R}_+$ defined by $|x|_v = \varepsilon^{v(x)}$ for each $x \in \mathbf{k}$.

The valuation v defined above is called an *additive valuation*. And an absolute value $|\cdot|$ on $\mathbf{k}$ is called a *multiplicative valuation*. In this note, the term *valuation* may refer to either an additive valuation or a multiplicative valuation, depending on the context.

Example 3. Let $\mathbf{k}$ be a field. The *trivial absolute value* on $\mathbf{k}$ is defined as

$$|x| := \begin{cases} 0, & x = 0; \\ 1, & x \neq 0. \end{cases}$$

Example 4. Let $|\cdot|_\infty$ be the usual absolute value on the field $\mathbb{Q}$ of rational numbers. Then $(\mathbb{Q}, |\cdot|_\infty)$ is a valuation field. Its completion is the field $\mathbb{R}$ of real numbers equipped with the usual absolute value.

Example 5. Let p be a prime number. The valuation v_p on $\mathbb{Q}$ induced by the ideal $(p) \subset \mathbb{Z}$ is called the *p-adic valuation*. It gives an absolute value $|\cdot|_p$ on $\mathbb{Q}$; see [Example 25](#)

Definition 6. The (*multiplicative*) *valuation group* of a valuation field $(\mathbf{k}, |\cdot|)$ is defined as the subgroup of $\mathbb{R}_{>0}$ given by

$$|\mathbf{k}^\times| := \{|x| : x \in \mathbf{k}^\times\}.$$

We use the notation $\sqrt{|\mathbf{k}^\times|}$ to denote the set $\{|x|^{1/n} : x \in \mathbf{k}^\times, n \in \mathbb{Z}_{>0}\}$.

Note that an absolute value $|\cdot|$ is non-trivial if and only if its valuation group $|\mathbf{k}^\times|$ is not equal to $\{1\}$.

Definition 7. Let $(\mathbf{k}, |\cdot|)$ be a valuation field. We say that $\mathbf{k}$ is *complete* if the metric $d(x, y) := |x - y|$ makes $\mathbf{k}$ a complete metric space.

Lemma 8. Let $(\mathbf{k}, |\cdot|)$ be a valuation field and $(\widehat{\mathbf{k}}, |\cdot|)$ its completion as a metric space. Then the operations of addition and multiplication on $\mathbf{k}$ can be extended to $\widehat{\mathbf{k}}$ uniquely, making $(\widehat{\mathbf{k}}, |\cdot|)$ a complete valuation field containing $\mathbf{k}$ as a dense subfield.

Proof. Note that the operations of addition and multiplication on $\mathbf{k}$ are uniformly continuous with respect to the metric $d(x, y) = |x - y|$. Thus they can be extended to $\widehat{\mathbf{k}}$ uniquely. $\square$

Proposition 9. Let $(\mathbf{k}, |\cdot|)$ be a complete valuation field with non-trivial absolute value. Then $\mathbf{k}$ is uncountable.

Proof. Since the absolute value $|\cdot|$ is non-trivial, we can construct a sequence $\{x_n\}_{n=1}^{\infty} \subseteq \mathbf{k}$ inductively such that $|x_n| < |x_{n-1}|/2$ for any $n \geq 1$ and $|x_0| < 1$. Then there is an injective map from $\mathbb{N}^{\{0,1\}}$ to $\mathbf{k}$ defined by

$$(a_n)_{n=1}^{\infty} \mapsto \sum_{n=1}^{\infty} a_n x_n, \quad a_n \in \{0, 1\}.$$

Since $|x_n| < 2^{-n}$, the series $\sum_{n=1}^{\infty} a_n x_n$ converges in $\mathbf{k}$. Note $|x_n| > |\sum_{m \geq n} x_m|$ for each n , we have that the map is injective. Thus $\mathbf{k}$ is uncountable. $\square$

Unlike the real number field $\mathbb{R}$, even a valuation field is complete, we can not expect the theorem of nested intervals to hold.

Definition 10. A valuation field $(\mathbf{k}, |\cdot|)$ is called *spherically complete* if every decreasing sequence of closed balls in $\mathbf{k}$ has a non-empty intersection.

Example 11. The field $\mathbb{C}_p$ of p -adic complex numbers is not spherically complete; see [Proposition 24](#).

Definition 12. Let $(\mathbf{k}, |\cdot|)$ be a valuation field. We say that $\mathbf{k}$ is *non-archimedean* if its absolute value $|\cdot|$ satisfies the *strong triangle inequality*:

$$|x + y| \leq \max\{|x|, |y|\}, \quad \forall x, y \in \mathbf{k}.$$

Otherwise, we say that $\mathbf{k}$ is *archimedean*.

Example 13. Let v be an additive valuation on a field $\mathbf{k}$. Then the induced absolute value $|\cdot|_v$ as in [Remark 2](#) is non-archimedean.

The converse is also true: if $(\mathbf{k}, |\cdot|)$ is a non-archimedean valuation field, then the function $v : \mathbf{k}^{\times} \rightarrow \mathbb{R}$ defined by $v(x) = -\log |x|$ is an additive valuation on $\mathbf{k}$.

The following proposition explains the terminology "archimedean".

Proposition 14. Let $(\mathbf{k}, |\cdot|)$ be a valuation field. Then $\mathbf{k}$ is archimedean if and only if the set $\{|n \cdot 1| : n \in \mathbb{Z}\}$ is unbounded.

Proof. If $\mathbf{k}$ is non-archimedean, then for any integer $n > 0$, we have

$$|n \cdot 1| = |1 + 1 + \cdots + 1| \leq \max\{|1|, |1|, \dots, |1|\} = 1.$$

Thus the set $\{|n \cdot 1| : n \in \mathbb{Z}\}$ is bounded.

Conversely, suppose that $\{|n \cdot 1| : n \in \mathbb{Z}\}$ is bounded by a real number $C > 0$. For any $x, y \in \mathbf{k}$, we have

$$|x + y|^n = \left| \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \right| \leq \sum_{k=0}^n C |x|^k |y|^{n-k}.$$

Without loss of generality, we assume that $|x| \leq 1$ and $y = 1$. Then the above inequality becomes

$$|x + 1|^n \leq \sum_{k=0}^n C |x|^k \leq C(n + 1).$$

Taking n -th root and letting $n \rightarrow \infty$, we obtain $|x + 1| \leq 1$. Hence $|x + y| \leq \max\{|x|, |y|\}$ for any $x, y \in \mathbf{k}$. $\square$

Theorem 15 (Ostrowski). Let $(\mathbf{k}, |\cdot|)$ be an archimedean complete valuation field. Then there exists an embedding $\iota : \mathbf{k} \hookrightarrow \mathbb{C}$ and $s \in (0, \infty)$ such that

$$|x| = |\iota(x)|_\infty^s, \quad \forall x \in \mathbf{k},$$

where $|\cdot|_\infty$ is the usual absolute value on $\mathbb{C}$. In other words, complete archimedean valuation fields are equivalent to either $\mathbb{R}$ or $\mathbb{C}$ equipped with usual absolute value (see [Definition 16](#)).

Proof. By [Proposition 14](#), $\mathbf{k}$ has characteristic zero. For any integers $n, m \in \mathbb{Z}_{>0}$, we write n in base m as

$$n = a_k m^k + a_{k-1} m^{k-1} + \cdots + a_1 \cdot m + a_0,$$

where $a_i \in \{0, 1, \dots, m-1\}$ for each i and $a_k \neq 0$. Then we have

$$|n| \leq \sum_i |a_i| |m|^i \leq k |m|^{k+1} \quad \text{and} \quad k \leq \frac{\log n}{\log m}.$$

Combining the above two inequalities, we obtain

$$\log |n| \leq \log(\log n) + \left(\frac{\log n}{\log m} + 1 \right) \log |m|.$$

Replace n by n^t , $t \in \mathbb{Z}_{>0}$, and divide both sides by t , we get

$$\log |n| \leq \frac{\log(\log n) + \log t}{t} + \left(\frac{\log n}{\log m} + \frac{1}{t} \right) \log |m|.$$

Let t trend to infinity, we get

$$\frac{\log |n|}{\log n} \leq \frac{\log |m|}{\log m}.$$

By symmetry, we have

$$\frac{\log |n|}{\log n} = \frac{\log |m|}{\log m}, \quad \forall n, m \in \mathbb{Z}_{>0}.$$

Set s be the constant $\log |n| / \log n$ for any $n > 1$. Then we have $|n| = n^s$ for any $n \in \mathbb{Z}$. By Proposition 14, we get $s > 0$. Without loss of generality, we may assume that $s = 1$ by replacing $|\cdot|$ by $|\cdot|^{1/s}$.

We have show that the natural embedding $\mathbb{Q} \hookrightarrow \mathbf{k}$ is an isometry if we equip $\mathbb{Q}$ with the usual absolute value $|\cdot|_\infty$. Thus it can be extended to an isometric embedding $\mathbb{R} \hookrightarrow \mathbf{k}$. For every $\alpha \in \mathbf{k}$, consider

$$f(z) := |\alpha^2 - (z + \bar{z})\alpha + z\bar{z}|, \quad z \in \mathbb{C}.$$

Since $\lim_{z \rightarrow \infty} f(z) = +\infty$, there exists $z_0 \in \mathbb{C}$ such that $f(z_0) = m$ is the minimum of f . Since $f^{-1}(m)$ is closed and bounded, this is a compact set; and then we can assume that z_0 has the largest norm among all points in $f^{-1}(m)$. We only need to show that $m = 0$.

Suppose the contrary that $m > 0$. Choose $0 < \varepsilon < m$. The polynomial $g(T) = T^2 - (z_0 + \bar{z}_0)T + z_0\bar{z}_0 + \varepsilon$ has two distinct roots $\beta, \bar{\beta} \in \mathbb{C}$. Then $\beta\bar{\beta} = z_0\bar{z}_0 + \varepsilon$ implies $|\beta| > |z_0|$; and hence $f(\beta) > m$.

Fix $n \in \mathbb{Z}_{>0}$, define

$$G(T) = (T^2 - (z_0 + \bar{z}_0)T + z_0\bar{z}_0)^n - (-\varepsilon)^n.$$

We have $G(\beta) = 0$ and $G(T) = g(T) \prod_{i=1}^{n-1} (T^2 - (\beta_i + \bar{\beta}_i)T + \beta_i\bar{\beta}_i)$ for some $\beta_i \in \mathbb{C}$. Then

$$f(\beta)m^{n-1} \leq |G(\alpha)| \leq |\alpha^2 - (z_0 + \bar{z}_0)\alpha + z_0\bar{z}_0|^n + \varepsilon^n = m^n + \varepsilon^n.$$

It follows that

$$f(\beta) \leq m + \frac{\varepsilon^n}{m^{n-1}}.$$

Let n trend to infinity, we get $f(\beta) \leq m$, a contradiction. $\square$

2 Places on a field

Definition 16. Let $\mathbf{k}$ be a field. Two absolute values $|\cdot|_1$ and $|\cdot|_2$ on $\mathbf{k}$ are said to be *equivalent* if there exists a real number $c \in (0, \infty)$ such that

$$|x|_1 = |x|_2^c, \quad \forall x \in \mathbf{k}.$$

Note that equivalent absolute values induce the same topology on the field $\mathbf{k}$. Moreover, the following lemma shows that the converse is also true.

Lemma 17. Let $\mathbf{k}$ be a field and $|\cdot|_1, |\cdot|_2$ be two absolute values on $\mathbf{k}$. Then the following statements are equivalent:

- (a) $|\cdot|_1$ and $|\cdot|_2$ are equivalent;
- (b) $|\cdot|_1$ and $|\cdot|_2$ induce the same topology on $\mathbf{k}$;
- (c) The unit disks $D_1 = \{x \in \mathbf{k} : |x|_1 < 1\}$ and $D_2 = \{x \in \mathbf{k} : |x|_2 < 1\}$ are the same.

Proof. The implications (a) $\Rightarrow$ (b) is obvious. Now we prove (b) $\Rightarrow$ (c). For any $x \in D_1$, we have $x^n \rightarrow 0$ as $n \rightarrow \infty$ under the absolute value $|\cdot|_1$ and thus under $|\cdot|_2$. Therefore, $|x|_2^n \rightarrow 0$ as $n \rightarrow \infty$,

which implies that $|x|_2 < 1$, i.e., $x \in D_2$. Similarly, we can prove that $D_2 \subseteq D_1$.

Finally, we prove (c) $\Rightarrow$ (a). If $|\cdot|_1$ is trivial, then $D_1 = \{0\}$ and thus $|\cdot|_2$ is also trivial. In this case, they are equivalent. Suppose that both $|\cdot|_1$ and $|\cdot|_2$ are non-trivial. Pick any $x, y \notin D_1 = D_2$. Then there exist real numbers $\alpha, \beta > 0$ such that $|x|_1 = |x|_2^\alpha$ and $|y|_1 = |y|_2^\beta$. Suppose the contrary that $\alpha \neq \beta$. Consider the domain $\Lambda \subseteq \mathbb{Z}^2$ defined by

$$\begin{cases} n \log |x|_2 < m \log |y|_2; \\ n\alpha \log |x|_2 > m\beta \log |y|_2. \end{cases}$$

Since $\alpha \neq \beta$, the two lines defined by the equalities are not parallel. Thus Λ is non-empty. Pick $(n, m) \in \Lambda$ and set $z := x^n y^{-m}$. Then we have $|z|_2 < 1$ and $|z|_1 > 1$, a contradiction. $\square$

Definition 18. Let $\mathbf{k}$ be a field. A *place* on $\mathbf{k}$ is an equivalence class of non-trivial absolute values on $\mathbf{k}$. An archimedean (resp. non-archimedean) place is also called an *finite* (resp. *infinite*) place. We denote the set of all places (resp. finite, infinite places) on $\mathbf{k}$ by $M_{\mathbf{k}}$ (resp. $M_{\mathbf{k}}^f, M_{\mathbf{k}}^\infty$). If $\mathbf{l} \subset \mathbf{k}$ is a subfield, we denote by $M_{\mathbf{k}/\mathbf{l}}$ the set of all places on $\mathbf{k}$ which are trivial on $\mathbf{l}$.

Example 19. Let $\mathbf{k} = \mathbb{C}(t)$ and $\mathbf{l} = \mathbb{C}$. Then I claim that

$$M_{\mathbb{C}(t)/\mathbb{C}} \cong \{ \text{prime divisors on } \mathbb{P}_{\mathbb{C}}^1 \}.$$

For each prime divisor P on $\mathbb{P}_{\mathbb{C}}^1$, we can define an additive valuation $\text{Mult}_P : \mathbb{C}(t)^\times \rightarrow \mathbb{Z}$ by assigning to each non-zero rational function $f \in \mathbb{C}(t)^\times$ its multiplicity at P . Fix a real number $\varepsilon \in (0, 1)$. Then we obtain a multiplicative valuation (absolute value) $|\cdot|_P$ on $\mathbb{C}(t)$ as in Remark 2. It is easy to check that the absolute value $|\cdot|_P$ is trivial on $\mathbb{C}$ and that different prime divisors give rise to inequivalent absolute values.

Conversely, given any non-trivial absolute value $|\cdot|$ on $\mathbb{C}(t)$ which is trivial on $\mathbb{C}$, by Proposition 14 and Example 13, the absolute value $|\cdot|$ is given by an additive valuation $v : \mathbb{C}(t)^\times \rightarrow \mathbb{R}$. Let $\mathcal{O}_v$ be the valuation ring of v and $\mathfrak{m}_v$ its maximal ideal. Then $t \in \mathcal{O}_v$ or $t^{-1} \in \mathcal{O}_v$. Without loss of generality, we assume that $t \in \mathcal{O}_v$. Since $|\cdot|$ is trivial on $\mathbb{C}$, we have $\mathbb{C}[t] \subseteq \mathcal{O}_v$. And we have $\mathfrak{m}_v \cap \mathbb{C}[t] \neq 0$ since otherwise v is trivial on $\mathbb{C}[t]^\times$ and thus on $\mathbb{C}(t)^\times$. It follows that the image of $\mathbb{C}[t]$ under the quotient map $\mathcal{O}_v \rightarrow \mathcal{O}_v/\mathfrak{m}_v$ is $\mathbb{C}$. This gives a point $P \in \mathbb{A}_{\mathbb{C}}^1 \subseteq \mathbb{P}_{\mathbb{C}}^1$. Then v is different from the additive valuation Mult_P by a positive scalar multiple via looking at the values on $\mathbb{C}[t]$. Thus we have established the claimed bijection.

Theorem 20 (Ostrowski). Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to either the usual absolute value $|\cdot|_\infty$ or a p -adic absolute value $|\cdot|_p$ for some prime number p . In other words,

$$M_{\mathbb{Q}} = \{|\cdot|_\infty\} \cup \{|\cdot|_p : p \text{ is a prime number}\}.$$

Proof. By Theorem 15, all archimedean places on $\mathbb{Q}$ are equivalent to the usual absolute value $|\cdot|_\infty$. Now we consider a non-archimedean absolute value $|\cdot|$ on $\mathbb{Q}$. By Example 13, $|\cdot|$ is induced by an additive valuation $v : \mathbb{Q}^\times \rightarrow \mathbb{R}$. Consider its valuation ring $\mathcal{O}_v$ and maximal ideal $\mathfrak{m}_v$. We have $\mathbb{Z} \subseteq \mathcal{O}_v$. If $\mathfrak{m}_v \cap \mathbb{Z} = 0$, then $\mathbb{Q} \subset \mathcal{O}_v$ and thus v is trivial, a contradiction. Thus there exists a prime number p such that $\mathfrak{m}_v \cap \mathbb{Z} = p\mathbb{Z}$. It follows that v is different from the p -adic valuation v_p by a positive scalar multiple via looking at the values on $\mathbb{Z}$. Hence $|\cdot|$ is equivalent to the p -adic

absolute value $|\cdot|_p$. □

Remark 21. For every non-archimedean place v on $\mathbb{Q}$ corresponding to a prime number p , we choose the unique normalized absolute value $|\cdot|_v$ in the class v such that $|p|_v = 1/p$. For the archimedean place v on $\mathbb{Q}$, we choose the usual absolute value $|\cdot|_v = |\cdot|_\infty$. Unless otherwise specified, we always use the normalized absolute values on $\mathbb{Q}$.

Remark 22. For any non-zero rational number $x \in \mathbb{Q}^\times$, one can easily check the *product formula*:

$$\prod_{v \in M_{\mathbb{Q}}} |x|_v = 1.$$

This can be viewed as an arithmetic analogue of the fact on $\mathbb{P}_{\mathbb{C}}^1$ that

$$\sum_{P \in \mathbb{P}^1(\mathbb{C})} \text{Mult}_P(f) = 0$$

for any non-zero rational function $f \in \mathbb{C}(t)^\times$. Indeed, fix a real number $\varepsilon \in (0, 1)$. Then by [Example 19](#), above fact can be rewritten as

$$\prod_{P \in \mathbb{P}_{\mathbb{C}}^1} |f|_P = 1.$$

Theorem 23 (Artin-Whaples approximations). Let $\mathbf{k}$ be a field and $v_1, v_2, \dots, v_n \in M_{\mathbf{k}}$ be pairwise distinct places on $\mathbf{k}$. For each $i \in \{1, 2, \dots, n\}$, let $x_i \in \mathbf{k}$ and $\varepsilon_i > 0$. Then there exists an element $x \in \mathbf{k}$ such that

$$|x - x_i|_{v_i} < \varepsilon_i, \quad \forall i \in \{1, 2, \dots, n\}.$$

In particular, the image of the diagonal embedding

$$\mathbf{k} \rightarrow \prod_{i=1}^n \mathbf{k}_{v_i}$$

is dense, where $\mathbf{k}_{v_i}$ is the completion of $\mathbf{k}$ with respect to the place v_i , and the product is equipped with the product topology.

Proof. We only need to find $c_i \in \mathbf{k}$ such that

$$|c_i|_{v_j} < 1, \quad \forall j \neq i; \quad \text{and} \quad |c_i|_{v_i} > 1.$$

If so, set $c_i(m) = c_i^m / (1 + c_i^m)$ for each $m \in \mathbb{Z}_{>0}$. Then we have

$$\lim_{m \rightarrow \infty} c_i(m) = \begin{cases} 0, & \text{under } v_j, j \neq i; \\ 1, & \text{under } v_i. \end{cases}$$

Thus the element $x = \sum_{i=1}^n c_i(m)x_i$ satisfies the required property for sufficiently large m .

Now we construct such $c = c_1$ by induction on n . By proof of [Lemma 17](#), there exists c with $|c|_1 > 1$ and $|c|_2 < 1$. This gives the base case $n = 2$. Suppose that we have constructed c for $n - 1$ places, i.e., $|c|_1 > 1$ and $|c|_i < 1$ for all $2 \leq i \leq n - 1$. There is $a \in \mathbf{k}$ such that $|a|_1 > 1$ and

$|a|_n < 1$. Consider

$$c_m = \begin{cases} \frac{c^m}{c^{m+1}}, & |c|_n > 1; \\ c^m, & |c|_n = 1; \end{cases}$$

we have

$$\lim_{m \rightarrow \infty} |ac_m|_i = \begin{cases} +\infty, & i = 1; \\ 0, & i = 2, \dots, n-1; \\ |a|_n < 1, & i = n. \end{cases}$$

Thus for sufficiently large m , the element $c = ac_m$ satisfies the required property. $\square$

Appendix

Proposition 24. The field $\mathbb{C}_p$ of p -adic complex numbers is not spherically complete.

Example 25. Let p be a prime number. For every $r \in \mathbb{Q}$, we can write r as $r = p^n \frac{a}{b}$, where $n \in \mathbb{Z}$ and $a, b \in \mathbb{Z}$ are integers not divisible by p . The p -adic absolute value on $\mathbb{Q}$ is defined as

$$|r|_p := p^{-n}.$$

The p -adic field $\mathbb{Q}_p$ can be described concretely as follows:

$$\mathbb{Q}_p = \left\{ \sum_{i=n}^{+\infty} a_i p^i \mid n \in \mathbb{Z}, a_i \in \{0, 1, \dots, p-1\} \right\}.$$

For $x = \sum_{i=n}^{+\infty} a_i p^i \in \mathbb{Q}_p$ with $a_n \neq 0$, its p -adic absolute value is given by $|x|_p = p^{-n}$. The operations of addition and multiplication on $\mathbb{Q}_p$ are defined similarly as those on decimal expansions.